

Toxoplasma gondii

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**



**World Health
Organization**

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Highlights

- There is strong evidence for transmission of *Toxoplasma gondii* through unsafe drinking-water, inadequate sanitation, and poor hygiene. *Toxoplasma gondii* is a significant zoonotic waterborne pathogen. Multiple drinking-water outbreaks of toxoplasmosis have been documented.
- Domestic cats and other felines are the definitive hosts of *T. gondii*. In the natural life cycle, felines acquire infection from the ingestion of infected rodents and birds.
- *Toxoplasma gondii* can be transmitted via two main routes: either by the faecal–oral route through ingestion of food or water contaminated with oocysts or via ingestion or consumption of tissue cysts in raw or undercooked meat from infected animals. Also, non-oral routes such as vertical transmission (mother-to-child), organ transplantation or blood transfusion can occur.
- Toxoplasmosis is usually asymptomatic in humans. If acute illness occurs, it is usually mild, with flu-like symptoms and enlargement of the lymph nodes, liver and spleen. Asymptomatic infection can result in ocular toxoplasmosis later in life, and other illnesses may develop in immunocompromised patients. After congenital infection, more severe outcomes such as miscarriage, stillbirth or fetal abnormality can occur, while ocular toxoplasmosis can also occur later in life, even after asymptomatic infection at birth.
- Toxoplasmosis affects people of all ages and in all income settings worldwide but is more prevalent in warm and humid climates. Poor socioeconomic status increases the risk of infection.
- The infective dose for *T. gondii* in humans has not been determined, although it is suspected that low doses of oocysts or tissue cysts may result in infection.
- *Toxoplasma gondii* oocysts can be detected in water by several methods, including microscopy and molecular techniques, but standard methods are not available.
- Limited numbers of studies have detected *T. gondii* oocysts in wastewater, surface water, groundwater, drinking-water, seawater and on crops. Oocysts can persist for long periods of time in water.
- Since felines are the definitive hosts of *T. gondii* systematic approaches that identify and control feline faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks from *T. gondii*.
- The reduction of *T. gondii* concentrations by wastewater and drinking-water treatment varies depending on the process. Removal is commonly based on size exclusion. *Toxoplasma gondii* is highly resistant to chemical disinfectants but is sensitive to UV light.
- Due to the long environmental persistence of *T. gondii* oocysts and their high resistance to chemical disinfectants, *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator for the presence or absence of *T. gondii* in drinking-water. Due to the long environmental persistence of *T. gondii* oocysts and their high resistance to chemical disinfectants, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator for the absence of *T. gondii* in drinking-water. However, the presence of *E. coli* is a suitable indicator of the potential presence of *T. gondii*.

1 General description

Toxoplasma gondii is a coccidian protozoan parasite in the phylum Apicomplexa. The existence of three genotypes (Types I, II, III) has been established based on virulence in mice. Type II and, to a lesser degree, Type III are predominant in Europe and North America. (Dardé, 2008; Fernández-Escobar et al., 2022). Atypical genotypes not belonging to the three main lineages are predominant in other continents (Dardé, 2008; Bahia-Oliveira et al., 2017). Domestic cats and other members of the Felidae family have been identified as the definitive host (Hill & Dubey, 2002). The *T. gondii* life cycle is complex and has two basic components: a sexual component that occurs only in felids and an asexual component that can occur in all warm-blooded animals. In the natural life cycle of the parasite, rodents and birds containing infective tissue cysts are eaten by felids. Sexual reproduction of the parasite occurs in the intestines of these animals and leads to the formation of oocysts (10 µm by 12 µm). Several million oocysts are shed in the faeces of cats for 7 to 21 days after acute infection (Dubey et al., 1970; Montoya & Liesenfeld, 2004). Depending on temperature and humidity in the natural environment oocysts require 1 to 5 days to sporulate and become infective. The asexual component in the host is an obligate, intracellular parasite (3 µm to 6 µm) called a tachyzoite. A chronic phase of the disease develops as tachyzoites transform into bradyzoites that form cysts in the host tissue (Gotteland et al., 2014; Bahia-Oliveira et al., 2017). Tissue cysts can persist for the duration of the host's life. *T. gondii* tissue cysts and oocysts excreted by cats are potentially infective when ingested by mammals (including man) and birds (Bahia-Oliveira et al., 2017).

2 Human health effects

Toxoplasmosis is usually asymptomatic in humans. In a small percentage of cases (~10%), flu-like symptoms and enlargement of the lymph nodes, liver and spleen present 5 to 23 days after the ingestion of cysts or oocysts (Montoya & Liesenfeld, 2004). During chronic infection, dormant tissue cysts persist in the host. In otherwise immunocompetent individuals, there is a risk of developing ocular toxoplasmosis later in life (Silveira et al., 2015; Gómez Marín et al., 2018), while individuals with severely suppressed immune systems (e.g. transplant or HIV/AIDS patients) are at risk of developing other diseases. Reactivation of tissue cysts can result in disseminated disease involving the central nervous system and lungs and lead to severe neurological disorders or pneumonia that can be fatal (Bahia-Oliveira et al., 2017). If a woman is infected for the first-time during pregnancy, the parasite is transmitted to the unborn child in about one third of cases. Infection during early pregnancy can lead to spontaneous abortion, stillbirth or foetal abnormality (Patel et al., 1996; Afonso et al., 2006; McLeod et al., 2006; Pappas et al., 2009). Congenital toxoplasmosis can result in chorioretinitis, cerebral calcifications, hydrocephalus, severe thrombocytopenia and convulsions when infection occurs in the first trimester or can remain asymptomatic when infection occurs late in pregnancy, but even in these cases, ocular toxoplasmosis can occur years after birth (McLeod et al., 2006).

3 Epidemiology

Toxoplasma gondii is found throughout the world, but infection is more common in warm and humid climates (Müller de Barros et al., 2022). The prevalence of human *T. gondii* infection

varies among countries and regions within the same country and has been reported with rates up to 90% (Molan et al., 2019). Poor socioeconomic status and the ingestion of raw or undercooked meat increase the risk of infection. Reports have indicated that the estimated seroprevalence of *T. gondii* infection in pregnant women varies widely in different parts of the world (Montoya & Liesenfeld, 2004; Hill et al., 2005). In addition, case-control studies in Europe have shown that consuming meat is the main risk factor for infection in pregnant women (Müller de Barros et al., 2022). In Ethiopia significant overall effect of anti-*Toxoplasma gondii* seropositivity among pregnant women was associated with having an unsafe water source as well as with age, abortion history, contact with cats, cat ownership, having knowledge about toxoplasmosis and being a housewife (Tarekegn et al., 2020).

4 Transmission

Toxoplasma gondii is transmitted via the consumption of raw and undercooked meat containing tissue cysts harbouring bradyzoites (Kijlstra & Jongert, 2009; Dubey, 2010) and through ingestion of oocysts in food, untreated or inadequately treated drinking-water or in surface waters during recreational bathing (Jones & Dubey, 2010; Gallas-Lindemann et al., 2013; Triviño-Valencia et al., 2016). In addition, ingestion of oocysts can also occur by consumption of contaminated vegetables and fruits and raw or undercooked shellfish (Jones et al., 2009; Putignani et al., 2011; Shapiro et al., 2015). Exposure to *T. gondii* oocysts through ingestion of soil (e.g. playing in sandboxes or gardening) is also possible (Bahia-Oliveira et al., 2017). Moreover, studies indicate that owning a cat is a risk factor for *T. gondii* infection (Jones et al., 2009).

Humans are an intermediate host and do not excrete oocysts; consequently, humans are not a source of transmission and are not considered contagious. Nevertheless, human-to-human transmission can occur through vertical transmission resulting in congenital toxoplasmosis (Gavinet et al., 1997; Elbez-Rubinstein et al., 2009; Bojar & Szymanska, 2010; Peyron et al., 2015), by organ transplantation or blood transfusion (Derouin et al., 2008; Foroutan-Rad et al., 2016).

The dose-response relationship for ingestion of *T. gondii* oocysts or tissue cysts has not been established in humans. However, it was shown that the dose-response relation for mice, rats and pigs is similar, and it was estimated that one oocyst is able to cause infection with a probability of 0.46 (Bonačić Marinović et al., 2020). Based on the animal data, the relative infectivity of *T. gondii* is expected to be high.¹

The incubation period for human hosts varies depending on the *T. gondii* life stage initiating infection. Reports from human outbreaks suggest incubation periods of 10 to 23 days after ingestion of bradyzoites in infected tissue cysts and 5 to 20 days after ingesting oocysts from contaminated environmental matrices (Meireles et al., 2015).

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be 1–10² organisms or particles, Moderate 10²–10⁴ and Low > 10⁴.

5 Detection methods

There is a lack of standard methods for *T. gondii* in environmental samples, which has hampered the capability to detect and quantify *T. gondii* in environmental media in many regions (Bahia-Oliveira et al., 2017). Methods used to detect *T. gondii* include microscopy, molecular techniques, loop-mediated isothermal amplification and bioassays (Dumètre & Dardé, 2003; Triviño-Valencia et al., 2016; Bahia-Oliveira et al., 2017). Microscopy and molecular methods do not discriminate between viable and non-viable oocysts.

6 Source and occurrence

6.1 Faecal pollution sources

Toxoplasma gondii oocysts have been found in sewage due to storm water runoff or households disposing cat faeces in the toilet. In a review of 62 raw sewage samples collected in four countries using a variety of detection methods, prevalence rates were 0% (Greece), 8.9% (Germany), 33% (Japan) and 43% (Bulgaria) (Bahia-Oliveira et al., 2017). In a more recent review, the presence of *T. gondii* in sewage ranged from 10% to 43% (López Ureña et al., 2022). Prevalence and persistence of oocysts in wastewater sludge or faecal sludge and treated wastewater are unknown.

6.2 Surface water

In a review of 2272 samples of surface waters in 10 countries using a variety of concentration and detection methods, prevalence rates of *T. gondii* were 0% (Germany and Japan), 5.9% (Iran), 7% (Pakistan), 6.7% and 7.7% (France), 8.8% (United Kingdom), 13% and 52% (Turkey), 12% and 64% (Bulgaria), 53% (Russian Federation) and 77% (Colombia) (Bahia-Oliveira et al., 2017). In a more recent review, the presence of *T. gondii* in surface water ranged from 5.0% to 77% (López Ureña et al., 2022). Oocysts of *T. gondii* persist in surface waters for long periods of time.² For example, Dumètre & Dardé (2003) reported that oocysts remain infectious in surface waters for 306 to 410 days on storage at 22.5 °C.

6.3 Groundwater

In a review of 175 groundwater samples in six countries using a variety of concentration and detection methods, prevalence rates of *T. gondii* were 0% (Germany and Japan), 7.5% (Pakistan), 14% and 50% (Bulgaria), 27% (Poland) and 100% (Russia) (Bahia-Oliveira et al., 2017). In a more recent review, the presence of *T. gondii* in groundwater ranged from 5.4% to 37% (López Ureña et al., 2022). Persistence in groundwaters is unknown but based on survival in surface waters it is expected to be long.

6.4 Drinking-water

Bahia-Oliveira et al. (2017) reviewed the presence of *T. gondii* in 160 drinking-water samples in five countries detected by a variety of concentration and detection methods. Prevalence rates

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: short, up to 1 week; moderate, 1 week to 1 month; long, over 1 month.

were 0% (Germany and Turkey), 3.3% (Pakistan), 8.3% and 33% (Bulgaria) and to 52% (Colombia). *Toxoplasma gondii* was also detected by mouse bioassay in an unfiltered drinking-water supply (Isaac-Renton et al., 1998). More recently, the presence of *T. gondii* DNA was detected in 2.3% of drinking-water samples in France, 8.8% of public water supply samples in Scotland and 100% of potable water samples in Brazil (López Ureña et al., 2022). Persistence of *T. gondii* oocysts in drinking-water is expected to be long, considering their survival in surface water and their ability to survive the disinfection procedures commonly used in drinking-water treatment.

6.5 Seawater

Limited studies are available on the detection of *Toxoplasma* in sea water. Koloren et al. (2013) detected *T. gondii* DNA in five of 14 seawater samples (36%).

Studies of filter-feeding bivalves demonstrated that marine ecosystems in four countries (Brazil, China, Turkey and the United States of America (USA) are contaminated with *T. gondii* (Bahia-Oliveira et al., 2017). Reports from five countries (Brazil, China, Italy, Turkey, USA) have confirmed the contamination of wild and/or commercially purchased seafood (Bahia-Oliveira et al., 2017). Due to the robust persistence of oocysts, *T. gondii* can remain viable for at least 24 months in seawater (Lindsay & Dubey, 2009), and retain infectivity for at least 85 days in oysters (Lindsay et al., 2004) and 3 days in mussels (Arkush et al., 2003).

6.6 Soil and crops

Soil contamination with *T. gondii* oocysts occurs after felids shed the parasite in their faeces (Gilot-Fromont et al., 2012). Distribution tends to concentrate in or near cat defecation sites (Gotteland et al., 2014). Nevertheless, oocyst dispersion occurs within the soil column or to other matrices by earthworms, arthropods, rain and wind (Dumètre & Dardé, 2003). *Toxoplasma gondii* has been detected in a variety of soils in several countries (Bahia-Oliveira et al., 2017; López Ureña et al., 2022). Prevalence ranged from 1.1% to 100% and 1.0% to 78% based on mouse bioassays and molecular methods, respectively. Contamination of crops may also occur due to the use of contaminated irrigation waters (López Ureña et al., 2022). Prevalence rates using molecular methods ranged from 8.2% to 33%.

Persistence and viability of oocysts in soil can be influenced by environmental factors such as temperature, humidity, vegetation and soil characteristics. Oocysts lose their ability to sporulate when exposed to extreme heat (50 °C for 10 minutes), extreme solar radiation or to freezing conditions (-21 °C for 1 day or -6 °C for 7 days). Once sporulated, oocysts are very resistant and can persist in moist soil for up to 18 months, even when exposed to temperatures ranging from -20 °C to 35 °C (Frenkel et al., 1975; Dumètre & Dardé, 2003). A study conducted in France reported a reduction of 1.13 log₁₀ of oocyst infectivity after 100 days in soil under dry conditions. In contrast, soil kept damp resulted in only 0.36 log₁₀ reduction of oocyst infectivity (Lelu et al., 2012).

Crops contaminated with *T. gondii* have been identified in various countries. Using microscopy oocysts were found in samples of a variety of raw vegetables; 1.9% in Pakistan (Haq et al., 2014), 4.9% in Egypt (Ahmad et al., 2016) and 6.6% in Saudi Arabia (Al-Megrin, 2010). Using molecular methods 0.26% of imported leafy greens in Canada (Lalonde & Gajadhar, 2016) and 9.7% of a variety of fruits and vegetables in Poland (Lass et al., 2012) contained *T. gondii*.

Under refrigeration, sporulated oocysts can remain infectious on berries for at least 8 weeks (Kniel et al., 2002).

7 Significance of sanitation

Since felids are the definitive host for *T. gondii*, a sanitation service chain that separates human faeces from the environment and from human contact will not interrupt transmission. A comprehensive approach that deals with the management of faeces from felines, primarily domestic cats is necessary to control the transmission of *T. gondii*.

8 Significance in drinking-water

Toxoplasma gondii is a significant zoonotic pathogen. There is strong evidence linking transmission of *T. gondii*, with unsafe drinking-water. The first documented waterborne toxoplasmosis outbreak linked to municipal drinking-water contamination occurred in Vancouver, Canada, where there was a sudden increase of acute toxoplasmosis and in the 100 individuals identified, 19% developed ocular toxoplasmosis (Bowie et al., 1997). The largest documented outbreak to date, was linked to the consumption of unfiltered water from a reservoir in Brazil and resulted in 290 illnesses (Keenihan et al., 2002; Vaudaux et al., 2010). Drinking-water was also identified as a source of outbreaks in Panama (Benenson et al., 1982), India (Hall et al., 1999), and the USA (Dubey et al., 1981; Krueger et al., 2014). The largest outbreak of ocular toxoplasmosis was described in Tamil Nadu in Southern India in 2004, which pointed to contaminated municipal drinking-water as the most plausible source (Balasundaram et al., 2010). Endemic toxoplasmosis is also prevalent in Brazil (Bahia-Oliveira et al., 2003) and Colombia (López-Castillo et al., 2005; Gómez-Marín, 2007) and has been linked to the consumption of unfiltered drinking-water.

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from *T. gondii*.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on physical removal (e.g. size exclusion by filtration through media or soil in septic tank drainage areas) or settling in wastewater treatment processes will reduce concentrations of protozoa such as *T. gondii*. Oxidizing chemical disinfectants, such as chlorine, are ineffective against *T. gondii* at the doses and contact times applied to drinking-water but UV light is effective at inactivating *T. gondii*.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat, discharge or reuse wastewater and faecal sludges that may contain cat faeces. However, data

are scarce on the effectiveness of centralized or on-site wastewater treatment systems to remove or inactivate *T. gondii* oocysts. Given their similarities to *Cryptosporidium* oocysts, it is anticipated that processes that control *Cryptosporidium* should also control *T. gondii*.

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023), control measures that can be applied to manage potential risks from *T. gondii* oocysts include prevention of source water contamination by wild and domestic cats and application of adequate water treatment processes and protection of drinking-water quality during storage and distribution.

Since *T. gondii* oocysts are larger than *Cryptosporidium* oocysts, filtration processes are important barriers for *T. gondii* in drinking-water treatment. Particle filtration, microfiltration and ultrafiltration systems are considered effective methods to remove oocysts (Bahia-Oliveira et al., 2017; Shapiro et al., 2019). *T. gondii* oocysts are highly resistant to chlorine.³ Wainwright et al. (2007a) reported that oocysts stored at 22 °C and pH 7.2 remained infectious after exposure to 100 mg/L free chlorine for up to 24 hours, making chlorine as applied in drinking-water and wastewater treatment ineffective against *T. gondii*. Other studies indicate that *T. gondii* oocysts are sensitive to UV light. The results of one study indicated that a 5 log₁₀ inactivation was achieved at a dose of 45 mJ/cm² (Wainwright et al., 2007b), while two other studies reported that doses of 10 and 15 mJ/cm² achieved a 3 log₁₀ inactivation (Dumètre et al., 2008; Ware et al., 2010).

Due to the long environmental persistence of *T. gondii* oocysts and their high resistance to chemical disinfectants, *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the possible presence or absence of *T. gondii* in drinking-water. Due to the long environmental persistence of *T. gondii* oocysts and their high resistance to chemical disinfectants, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator for the absence of *T. gondii* in drinking-water. However, the presence of *E. coli* is a suitable indicator of the potential presence of *T. gondii*.

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

10 ICD-11 classification

XN7HC	<i>Toxoplasma</i>
XN896	<i>Toxoplasma gondii</i>
1F57	Toxoplasmosis

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.